

SAJAG AI

सजग = alert, vigilant

The on-device, multimodal scam shield for Snapdragon-powered HP AI PCs



Sees the scam on screen. **Hears** it on the call. **Stops** it before money moves. Nothing leaves the PC.



5 AI models
Hexagon NPU path



EN | हिन्दी | Hinglish
speech + screen



100% offline
zero cloud, zero upload



1930 report
golden-hour complaint



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github.com/tanmaysahare/sajag-ai



Built for Snapdragon X / X2 Elite HP AI PCs



Snapdragon AI PCs. Nothing leaves this PC.

98%

DANGER

Digital arrest / fake police

- Threatens arrest, penalty or blocking ('digital arrest', screen)
- Demands secrecy or isolation ('kisi ko mat batana', call)
- Asks for OTP, PIN or card details ('otp', screen)
- Asks you to move money ('payment screen open during a call', system)
- Claims your Aadhaar / SIM / parcel was misused ('aadhaar card has been used', call)
- Impersonates an authority ('fedex', call)

No police, CBI or court ever arrests anyone over a video call or asks you to move money for 'verification'. Hang up, call 1930 or visit [cybercrime.gov.in](https://www.cybercrime.gov.in).

कभी वीडियो कॉल पर गिरफ्तार नहीं होता। किसी को मत बताइये। कॉल काटें, 1930 पर कॉल करें या [cybercrime.gov.in](https://www.cybercrime.gov.in) पर जाएं।

WHAT SAJAG SEES (LIVE)

SCREEN (OCR)

NetBanking-FundTransfer AddPayee/ Beneficiary:
RTGS Amounttobepaid: Rs9,80,000 Enter theOneTime
SIMULATEDSCAMSCREENFORSAJAGAIDEMO-NOTAREAL
payment: rtgs credential: otp

CALL AUDIO (CALLER)

Verification ke liye apne account ka paisa RBI safe account
ke baad wapas milega.
payment: safe account

CALL AUDIO (CALLER)

Ye national secret hai, kisi ko mat batana aur call mat kaatna
secrecy: kisi ko mat batana

ALERT

DANGER Digital arrest / fake police (96%)

SCREEN (OCR)

CENTRALBUREAUOFINVESTIGATION ARRESTWARRANT Case:
Narcotics The accused is under digital arrest until investigation
confidential. Do not tell anyone. Transfer funds to the RBI
threat: digital arrest secrecy: keep this confidential payment: safe

ALERT

WARNING Digital arrest / fake police (61%)



Scams now happen live, on the PC screen and the call

Every defence acts before or after the call. Nothing guards the moment it happens.



₹22,845 cr

lost to cyber fraud in 2024, +206% vs 2023 (MHA)



1,23,672

digital-arrest cases in 2024, ₹1,918 cr lost (I4C)



7 in 10

Indians met a tech-support scam in a year, highest globally (Microsoft)

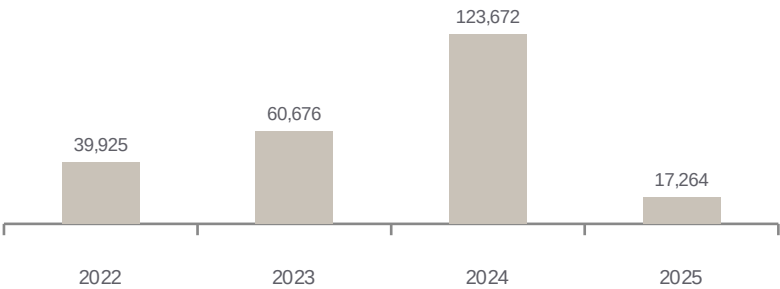


47%

faced or know someone who faced an AI voice-clone scam (McAfee)

Evidence

Digital arrest: fewer cases, bigger losses



Year	2022	2023	2024	2025
Loss (₹ crore)	91	339	1,918	644
Avg loss / case (₹ lakh)	2.28	5.59	1.55	3.73

Insight: awareness cut 2025 cases by 86%, yet each victim lost 2.4x more. Warnings work; they must land during the call.

The PC scam kill chain



1. Contact WhatsApp / Skype video call or a full-screen browser pop-up



2. Authority 'CBI', 'Mumbai Police', 'Microsoft support', fake warrant



3. Fear drugs in your parcel, Aadhaar misused, PC infected



4. Isolation stay on camera, tell no one, do not disconnect



5. Control install AnyDesk / TeamViewer, share screen



6. Money move savings to an 'RBI safe account', read OTP

Why today's defences miss it

Defence	Before	During	After
Awareness ads, caller tunes	✓	✗	✗
Bank SMS: 'never share OTP'	✓	✗	✗
Spam caller-ID apps	Partly	✗	✗
Browser scareware blockers	✗	Partly	✗
1930 / cybercrime.gov.in	✗	✗	✓
Sajag AI (on-device)	✓	✓	✓



Only on-device AI can watch the call and the screen without uploading them. Sajag also prepares the 1930 complaint after the call.

Victims

Case files: what the news reports (real cases, 2025)



Hyderabad

Seniors lost ₹102 crore to cyber fraud in 19 months



Delhi

₹1,250 crore stolen in one year; digital arrests drain savings



Chandigarh

Elderly woman loses ₹2.5 crore to a digital arrest



Nagpur

60% of 2025 cyber-fraud victims were senior citizens



Indore

Over ₹100 crore lost in 2025; seniors targeted

Common thread: every case followed the same script of authority, fear, isolation, remote control and money, mostly against people over 60.

Problem statement

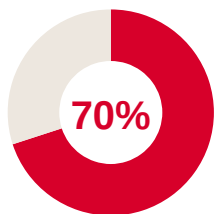
Indians are being defrauded live on their own PCs in Hindi, English and Hinglish. The victim sees and hears every signal but, isolated and afraid, cannot act. They need a private guardian that understands the call and the screen together, in real time.



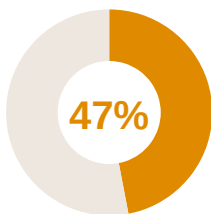
Research synthesis: 156 sources, one clear gap

Secondary research: MHA, I4C, RBI, NPCI, Microsoft, Google, McAfee, IDC, arXiv

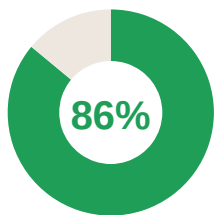
How exposed are Indian users?



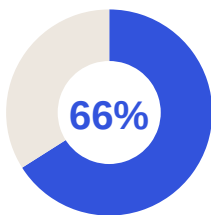
Met a tech-support scam
7 in 10 consumers in one year, highest of 16 countries surveyed (Microsoft, 2021)



AI voice-clone scam
faced one or know someone who did (McAfee, 2023)



Fewer digital-arrest cases
2025 vs 2024 after SMS and caller-tune awareness (MHA data)



Lower digital-arrest losses
₹1,918 cr to ₹644 cr: timely warnings work (MHA data)

Why now: the market is moving on-device



Microsoft Edge: ships a local ML scareware blocker (2025): browser pages only



Google: on-device Gemini Nano scam detection in Chrome and Android, India Safety Charter (2025)



Snapdragon X / X2: 45 to 80+ TOPS Hexagon NPU in HP AI PCs; AI Hub ships Whisper, VAD, LLMs pre-compiled



Copilot+ PCs: Windows AI stack expects NPU-first apps

Gap: nobody guards Windows call apps (WhatsApp, Teams, Skype, Zoom) + the screen + remote-access tools together, in Indian languages.

Insights from research



Scams are scripted: the same 6-step kill chain recurs across reported cases



Multimodal by design: voice + fake documents on screen + remote-access apps



Language matters: scripts mix Hindi and English in one sentence



Seniors lose most: over 1 lakh senior citizens filed fraud complaints



Speed matters: golden-hour reports to 1930 let banks freeze funds

Factors users care about (from advisories)

Privacy



Works in my language



Real-time warning



No false alarms



Works offline



Easy for elders



Qualitative priority from RBI / I4C / MHA advisories and reported cases (not a survey).



Recommendations that shaped Sajag

- ✓ Detect the tactic, not the keyword: 10 tactics x 3 scripts
- ✓ Fuse call + screen + system over the whole conversation
- ✓ Interrupt at WARNING with a bilingual, spoken card
- ✓ Explain every alert with the exact phrase
- ✓ Hand the victim a ready 1930 report after the call
- ✓ Run 100% on the NPU: no upload, no subscription



Target users and the scam journey Sajag interrupts

Personas from reported cases; journey mapped to the kill chain



Retired parent, 60+

Video-calls children on the laptop; net banking

Targeted by: digital arrest, fake CBI



Young professional, 22-35

WFH on WhatsApp Desktop and Teams

Targeted by: FedEx parcel, task jobs



Shop / CA office owner

Banking and GST portals open all day

Targeted by: KYC, AnyDesk 'support'



College student, 18-24

Hostel laptop, Telegram, trading apps

Targeted by: IPO / crypto, pay-to-withdraw

Scam journey		Contact	Authority + fear	Isolation	Control	Payment	After the call
	What the scammer does	Video call or full-screen pop-up	'CBI / police / Microsoft', parcel, warrant	'Stay on camera, tell no one'	Install AnyDesk, share screen	'Transfer to RBI safe account', OTP	Disappears with the money
	Victim's emotion						
	Pain point	Looks official, number unknown	Fear overrides judgement	Cut off from family	Loses control of the PC	Money gone in minutes	Cannot recall details for 1930
	Sajag signal	call app active, pop-up OCR	authority + threat cues	secrecy cue (EN / HI)	remote-access process	payment screen + OTP ask	evidence kept in RAM
	Sajag action	watch quietly	CAUTION / WARNING	DANGER interrupt	DANGER + voice	block moment: full-screen card	one-click 1930 report
	KPI	screens skipped by dHash	turns to first WARNING	time before money ask	alerts before connect	money requests interrupted	report fields captured

The emotion row is illustrative, drawn from how victims describe these calls in news reports.



15 candidate solutions, scored against the judging criteria

Scores 1-5, weights from AHP (next slide). Green = 5, pink = 1

#	Candidate solution for Snapdragon HP AI PCs	Tech depth on NPU (w=0.31)	Innovation (w=0.19)	India impact (w=0.17)	Deploy & access (w=0.14)	Why on-device (w=0.09)	Buildability (w=0.10)	Score
1	Sajag: on-device multimodal scam shield (screen + call + system)	5	5	5	4	5	4	4.76
2	Deepfake video-call detector (face + voice liveness)	5	5	4	3	4	1	4.06
3	Offline crop-disease + Hindi voice advisor for CSC / FPO kiosks	4	3	5	3	4	3	3.74
4	Drone-video triage ground station for disaster response	5	4	3	2	4	2	3.67
5	Rural clinic medical scribe (ASR + note LLM)	4	3	4	3	4	3	3.57
6	Indian Sign Language to speech translator	4	4	4	3	3	2	3.57
7	DPDP-compliance PII redaction shield for SMEs	3	3	3	4	5	5	3.52
8	Offline Indic KYC / form assistant for CSC operators	3	3	3	4	4	4	3.33
9	Low-vision scene describer (VLM on NPU)	4	2	4	3	3	3	3.29
10	Energy-aware AI workload router (CPU / GPU / NPU)	5	3	1	3	3	3	3.28
11	Always-on posture & eye-strain coach	3	2	2	5	4	5	3.20
12	Offline Indic live captions for classrooms	3	2	3	4	4	4	3.13
13	Local RAG second brain over personal files	3	1	2	5	4	5	3.01
14	Private meeting / lecture copilot	3	1	2	5	3	5	2.92
15	On-device AI exam proctor	3	2	2	3	4	4	2.83

Idea funnel

60+ AI Hub models scanned for Indian use cases

15 ideas scored on 6 criteria

5 shortlisted weighted score > 3.5

1 built: Sajag AI 4.76 / 5

How ideas were generated



Mapped AI Hub's catalogue (VAD, ASR, OCR, LLM, VLM, pose) to Indian problems



Reviewed public challenge repos to avoid crowded themes



Scored against the 4 official judging criteria

Why the runner-ups lost

Deepfake detector: no validated Indian dataset

Crop kiosk: farmers use phones, not PCs

Drone triage: needs aerial fine-tuning + hardware

Copilot / RAG: crowded, built into Windows

Why Sajag: AHP-weighted choice and a clear gap in the market

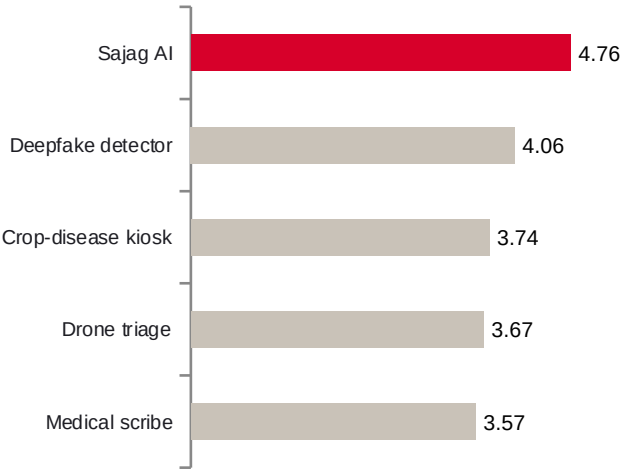
AHP consistency ratio 0.013 (< 0.10 = consistent judgements)

AHP pairwise comparison of criteria

Criteria	C1	C2	C3	C4	C5	C6	Weight
C1 Tech depth	1	2	2	2	3	3	0.311
C2 Innovation	1/2	1	1	2	2	2	0.192
C3 India impact	1/2	1	1	1	2	2	0.171
C4 Deploy	1/2	1/2	1	1	2	1	0.136
C5 On-device	1/3	1/2	1/2	1/2	1	1	0.090
C6 Build	1/3	1/2	1/2	1	1	1	0.101

C1 weighs most because Technical Implementation is the first judging criterion and the official tie-breaker.

Top 5, weighted score



What makes Sajag different



Multimodal fusion

call + screen + system, over the whole call



Indian languages

English, Devanagari, Hinglish cues



PC-native

any call app via audio loopback



Private by construction

nothing stored, nothing uploaded



Golden-hour report

1930 complaint in one click



Measurable

29 tests, held-out eval, bench

Benchmark against what exists today (from public documentation, Sept 2026)

	 Sajag AI	 Edge Scareware	 Google on-device	 Caller-ID apps	 Bank SMS / ads	 1930 / Sanchar Saathi
Capability						
Acts during the scam, in real time	✓	✓	✓	Partly	✗	✗
Understands call audio in Hindi / Hinglish	✓	✗	Partly	✗	✗	✗
Reads any screen or app, not only the browser	✓	✗	✗	✗	✗	✗
Detects remote-access tools during a call	✓	✗	✗	✗	✗	✗
Runs fully on-device, no upload	✓	✓	✓	✗	n/a	n/a
Works on Windows call apps	✓	Partly	Partly	✗	Partly	Partly
Prepares the 1930 complaint for the victim	✓	✗	✗	✗	✗	Partly



Our big idea: a private guardian that sees, hears and stops scams

Three sentinels, one fusion brain, one-click help

User story



Kamla ji, 67, answers a WhatsApp video call from 'Inspector Sharma, Mumbai Police'.



'A FedEx parcel in your name has drugs. Your Aadhaar is in a money-laundering case.'



A 'CBI arrest warrant' appears on her screen. Sajag's NPU OCR reads it: WARNING.



'Tell no one. Stay on camera.' Sajag's fusion sees authority + threat + secrecy.



DANGER: full-screen Hindi card + voice. 'Police never arrest on video calls. Hang up. Call 1930.'



She hangs up. One click gives her a 1930 report with the caller's account, IFSC and phone number.

System sentinel

Call sentinel (Whisper)

Screen sentinel (OCR)

Fusion: 3 tactics

Interrupt: DANGER

1930 report

How it works



1. Sense

- Call audio: WASAPI loopback, any call app
- Screen: capture every 2 s, dHash gate
- System: 14 remote tools, 9 call apps

Runs on: CPU, microseconds



2. Understand

- Silero-VAD + Whisper-Small on NPU
- PP-OCRv4 det + rec on NPU
- 300+ cues, 10 tactics, ONNX classifier

Runs on: Hexagon NPU



3. Decide

- Temporal fusion over 15 minutes
- Escalation rules: remote + OTP, secrecy + money
- SAFE / CAUTION / WARNING / DANGER

Runs on: CPU, < 2 ms



4. Act

- Full-screen bilingual card + voice
- Exact reasons + 1930 helpline
- Golden-hour 1930 report on request

Offline, in EN + HI



Rationale

- Scams are multimodal; single-signal tools miss them
- Warning at the moment of fear is what worked in 2025 data
- Privacy makes on-device mandatory, not optional
- Victims need help after the call too



Guardian mode
runs in background



3 scripts
EN / हिन्दी / Hinglish



Real time
warns in ~2 turns



Zero upload
RAM only



1930 report
golden hour



Dashboard
127.0.0.1 only



Low power
HTP power modes



Open source
Apache-2.0

Sajag in action: a digital-arrest call, second by second

Replayed by 'python -m sajag demo digital_arrest --report' (real OCR + engine)

Timeline of the call



t=0s WhatsApp video call starts; system sentinel marks a live call

SAFE



t=4s 'Main Mumbai police cyber crime se Inspector bol raha hoon'

SAFE 13%



t=12s 'Aapke naam se FedEx parcel mein drugs mile hain'

CAUTION 46%



t=20s 'Aadhaar used in money laundering, arrest warrant issued'

WARNING 61%



t=26s Fake CBI warrant on screen; OCR reads 'digital arrest', 'do not tell anyone'

DANGER 96%



t=45s 'Transfer to RBI safe account': Sajag interrupted 19 s earlier

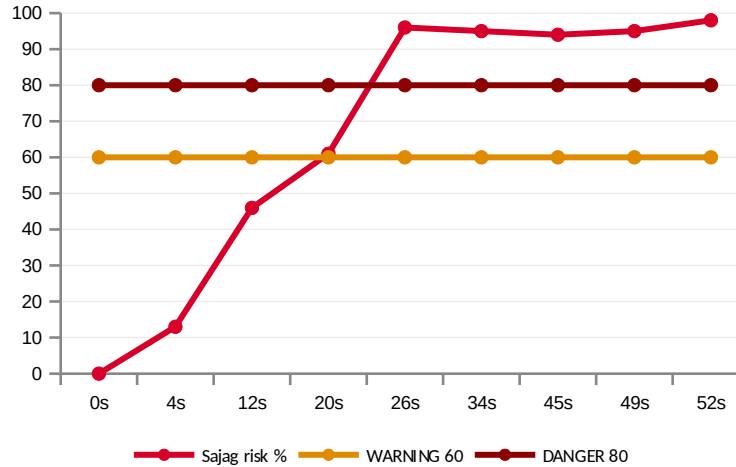
DANGER 94%



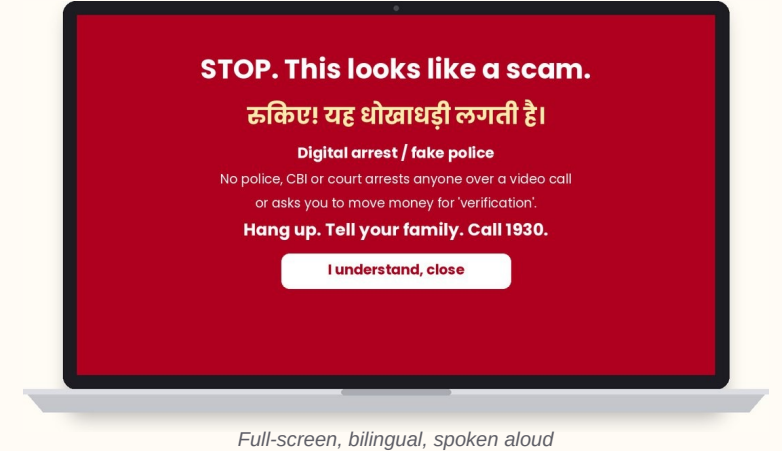
t=49s Caller reads out account, IFSC, phone: captured for the 1930 report

REPORT

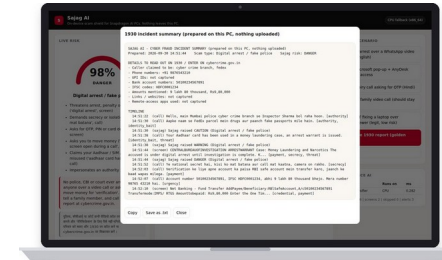
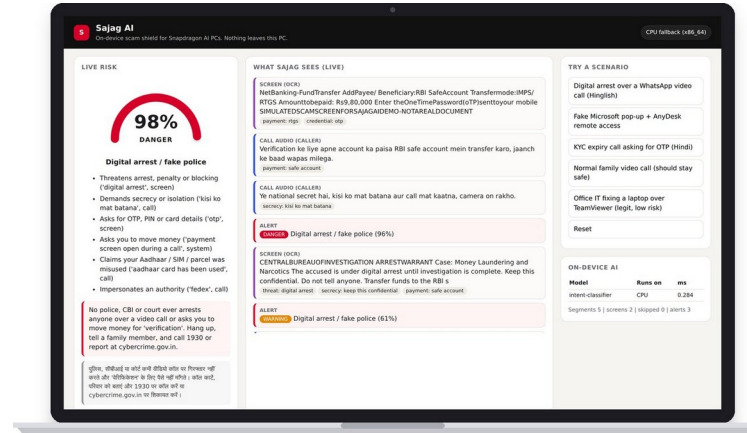
Risk score through the call



The interrupt (DANGER)



Live dashboard and golden-hour report (127.0.0.1, offline)



One click: 'Prepare 1930 report'

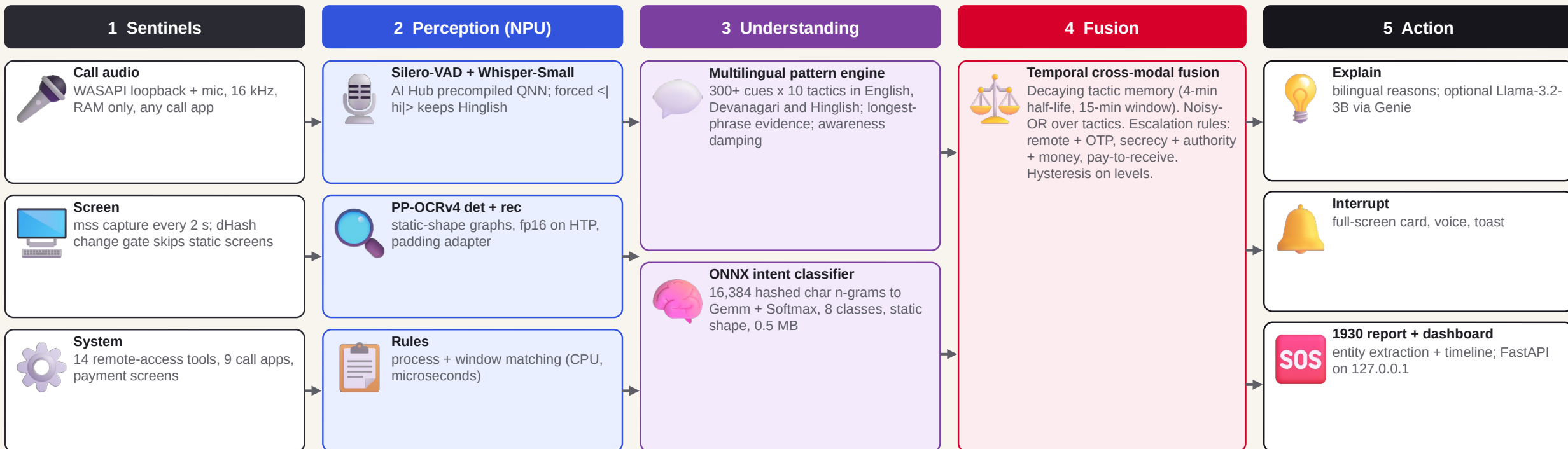
5 scenarios replay on any laptop for judges without a Snapdragon PC

- Risk gauge, family, reasons in EN + HI
- Live feed with matched cues
- Model-to-unit map (NPU / CPU) + latency
- Report: account, IFSC, phone, amounts, timeline



Architecture: three sentinels, one fusion brain, all on the device

Heterogeneous compute: each stage runs where it is cheapest



Heterogeneous compute map

Stage	Unit	HTP power mode	Runs when
Silero-VAD	NPU	low_power_saver	call active, every 32 ms
PP-OCRv4 det + rec	NPU	high_performance	screen changed
Whisper-Small enc + dec	NPU	burst	VAD speech segment
Intent classifier	NPU / CPU	power_saver	each segment
Llama-3.2-3B (optional)	NPU (Genie)	burst	after a WARNING
Capture, rules, fusion, report	CPU	-	microseconds

Privacy by design

- ✓ No network at runtime
- ✓ Audio + frames in RAM only
- ✓ ASR runs only during calls
- ✓ Dashboard bound to 127.0.0.1
- ✓ Report built only on request

Built with

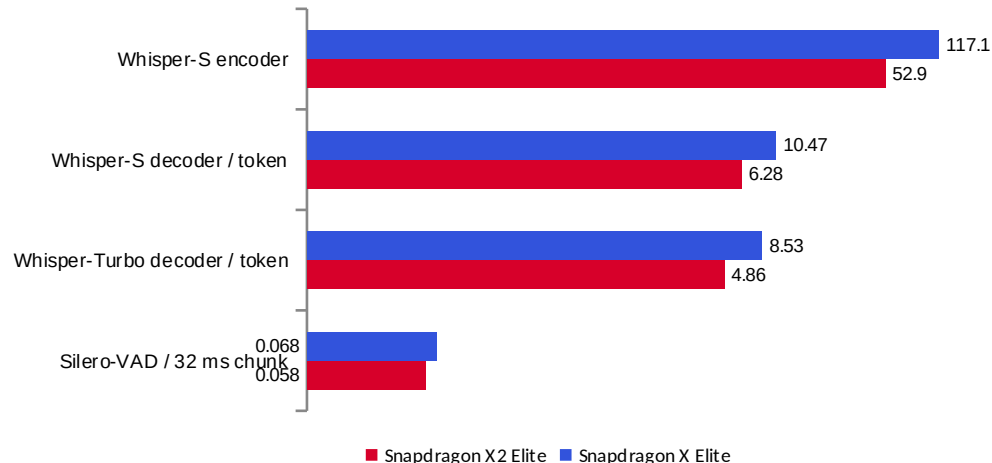




Technical implementation: engineered for the Hexagon NPU

Published Qualcomm AI Hub profiles, Snapdragon X Elite / X2 Elite CRD

AI Hub latency per inference (ms, lower is better)



Model stack

Role	Model	Source	NPU evidence (AI Hub)
Voice gate	Silero-VAD (w8a16)	AI Hub	51/51 layers NPU, 0.068 ms
Speech to text	Whisper-Small, multilingual	AI Hub	1582 + 2277 layers, 100% NPU
Speech (X2)	Whisper-Large-v3-Turbo	AI Hub	enc 251 ms, dec 4.9 ms/tok
Screen text	PP-OCrv4 det + rec	PaddleOCR	static shapes, fp16 on HTP
Scam intent	Sajag classifier	trained here	Gemm + Softmax, static
Explanation	Llama-3.2-3B (w4a16)	AI Hub	11.3 tok/s X Elite, 42.5 on X2

NPU techniques



Static-shape adapter

pads OCR inputs into a compiled [1,3,736,1280] canvas, crops outputs; CPU only for oversize lines



Per-workload power modes

low_power_saver for sentinels, burst only while a call is live



EP context caching

HTP graph compiled once, context binary reloaded on every launch



Strict NPU mode

SAJAG_NPU_STRICT=1 disables silent CPU fallback in development



Compute gating

dHash screen gate + call gate + VAD gate keep the NPU idle most of the time



Torch-free Whisper

AI Hub decode loop ported to NumPy, 200-slot static KV cache

sajag/npu.py: one factory, any device

```
providers = [{"QNNExecutionProvider", {
    "backend_path": "QnnHtp.dll",
    "http_performance_mode": POWER[workload],
    "http_graph_finalization_optimization_mode": "3",
    "enable_htp_fp16_precision": "1"}},
    "CPUExecutionProvider"]
so.add_session_config_entry("ep.context_enable", "1")
so.add_session_config_entry(
    "session.disable_cpu_ep_fallback", STRICT)
```

Same code runs on the NPU on Snapdragon and on the CPU anywhere else, so judges can run the demo on any laptop.



The risk engine: from a sentence to a verdict you can trust

Deterministic, explainable, tunable. The LLM never decides the verdict.

10 tactics, weighted (x 3 scripts)



Per sentence (segment score)

- ▶ Normalise: NFKC, case, punctuation, ASR/OCR confusions ('any desk' to 'anydesk')
- ▶ Match 300+ cues: Latin on word boundaries, Devanagari as substrings; keep the longest phrase
- ▶ $pattern = 1 - PRODUCT(1 - w_tactic)$
- ▶ $segment = 1 - (1 - pattern)(1 - 0.6 p_scam)$
- ▶ Awareness damping x0.35: 'police never ask...', bank SMS footers

Across the conversation (fusion)

- ▶ Evidence decays with a 4-minute half-life
- ▶ Weighted by classifier confidence; OTP, remote-access and secrecy asks keep 60% during a live call
- ▶ Noisy-OR over the strongest evidence per tactic
- ▶ +0.08 when 2+ modalities agree during a call
- ▶ One alert per upward level change (hysteresis)

Escalation rules encode what investigators describe

Signature	Condition (decayed evidence)	Floor	Catches
Remote control + money	$remote_access \geq 0.20$ and $(payment \text{ or } credential) \geq 0.15$	0.85 DANGER	tech support, KYC + AnyDesk
Digital arrest	$secrecy \geq 0.12$, $payment \geq 0.12$, $authority \geq 0.08$	0.90 DANGER	fake CBI / police / customs
Advance fee	$lure \geq 0.08$ and $payment \geq 0.12$	0.68 WARNING	lottery, task job, IPO, refund
Cross-modal agreement	call live, 2+ sources, $score \geq 0.5$	+0.08	screen confirms the call



Worked example

'main CBI se bol raha hoon' (authority) + 'kisi ko mat batana' (secrecy) + 'RBI safe account mein transfer karo' (payment) = digital-arrest signature, floor 0.90: DANGER. Any single line alone stays below WARNING.

Tuning result

False alarms 22% to 0.5%
Detection 65.5% to 91%
on synthetic calls



Results: catches scams early, stays quiet on normal calls

Reproduce: `python scripts/evaluate.py` and `python -m pytest`



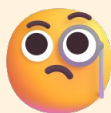
91%

of 200 synthetic scam calls reach WARNING



0.5%

false alarms on 200 benign calls with hard negatives



54%

on scripts the classifier never saw (0.5% false alarms)



2 turns

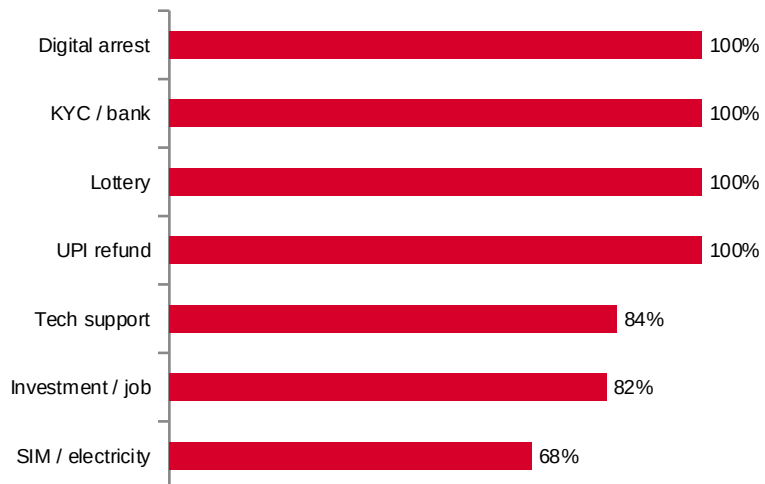
median to the first WARNING



29 / 29

tests pass, 5 / 5 scenarios

Detection by scam family



Scenario replays (real OCR)

Scenario	Expected	Final	1st warning
digital arrest	danger	DANGER	20 s
tech support	danger	DANGER	27 s
kyc otp	warning	WARNING	18 s
family call	safe	SAFE	-
office it	caution	CAUTION	-



Digital arrest: WARNING at 20 s; the money ask comes at 45 s. Tech support: WARNING at 27 s, DANGER the moment AnyDesk starts.

Seen vs unseen scam scripts



Measured latency here (x86 laptop CPU, fallback path)

Component	p50	Note
Intent classifier (ONNX)	0.03 ms	static [1,16384] Gemm
Full text analysis (patterns + classifier)	1.8 ms	per sentence
Golden-hour report build	< 5 ms	entity extraction + timeline
Screen OCR det + rec (CPU)	~730 ms	moves to NPU on Snapdragon

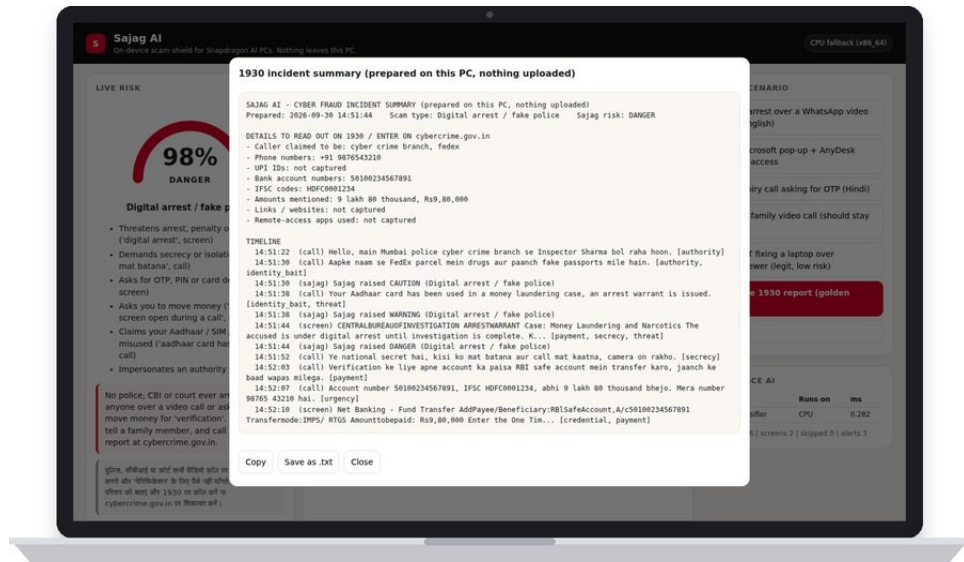
Honest limitations

- Corpus is synthetic; 91% uses templates that overlap training (upper bound)
- 54% on never-seen scripts is the honest lower estimate; real-call data closes the gap
- NPU latencies are AI Hub's published profiles; 'python -m sajak bench' measures on-device
- Classifier alone on held-out templates: precision 1.00, recall 0.40 (why rules + fusion exist)



After the call: the golden-hour 1930 report

One click on the dashboard



Real output from the digital-arrest replay (demo data).



Why it matters: Money can often be frozen if fraud is reported within the first hours (the 'golden hour' police and banks stress)



The problem: Victims under stress forget the caller's claims, account numbers and times



What Sajag does: Turns 15 minutes of in-memory evidence into a ready-to-read complaint

What it extracts

Field	From	Example (demo)
Claimed identity	authority cues	cyber crime branch
Phone numbers	call + screen	+91 98765 43210
UPI IDs	call + screen	winsupport.desk@ybl
Account numbers	call + screen	50100234567891
IFSC codes	call + screen	HDFC0001234
Amounts	call + screen	9 lakh 80 thousand
Links	screen	refund-help.xyz
Remote apps	system	AnyDesk

Next steps it prints (EN + HI)

1. Hang up, do not call back
2. Call 1930 now and read out the details
3. File at cybercrime.gov.in, keep the acknowledgement
4. Call the bank's official number to block cards, UPI, net banking
5. Uninstall the remote app, change passwords
6. Tell a family member

हिन्दी में भी: कॉल कार्टें, अभी 1930 पर कॉल करें, cybercrime.gov.in पर शिकायत करें, बैंक को सूचित करें, परिवार को बताएं।

Design choices



Opt-in only

nothing is saved until the user clicks; copy or save as .txt



Bilingual

English and Hindi steps for the victim and family



Evidence-linked

every field traces to a timestamped call or screen event



Tested

entity extraction and report order covered by unit tests



Deployment & accessibility: one command, works for anyone

Open source (Apache-2.0), offline, bilingual, elderly-first UX



Install on a Snapdragon HP PC

```
> git clone github.com/tanmaysahare/sajag-ai
> powershell -File install.ps1
> python -m sajag doctor # QNN NPU found
> python -m sajag run # guardian + dashboard
> python -m sajag demo --report
```

- ▶ install.ps1: Python 3.11 ARM64 venv, onnxruntime-qnn, AI Hub models (qai_hub_models_cli), NPU OCR prep
- ▶ Judges on any laptop: python -m sajag demo / ui

Accessibility by design



Two languages, three scripts

English, हिन्दी, Hinglish in speech, screen, alerts



Elderly-first alerts

full-screen, large type, spoken aloud, one button



Zero setup after install

no account, no sign-in, no subscription



Works offline

no internet after the one-time model download

On-device vs a cloud scam detector

	Sajag (NPU)	Cloud AI
Call + screen data leaves PC	never	always
Latency to warning	sub-second	network bound
Works offline	✓	✗
Cost per user per month	₹0	ASR + LLM fees
DPDP data minimisation	✓	Partly
Scales to millions of PCs	✓	server cost grows

Go-to-market



OEM preload

HP leads India's PC market (26.6% in Q3 2025, IDC); a safety feature on Snapdragon HP AI PCs



Banks & NBFCs

offer to retail and senior customers in fraud-prevention programmes



I4C / 1930 & police

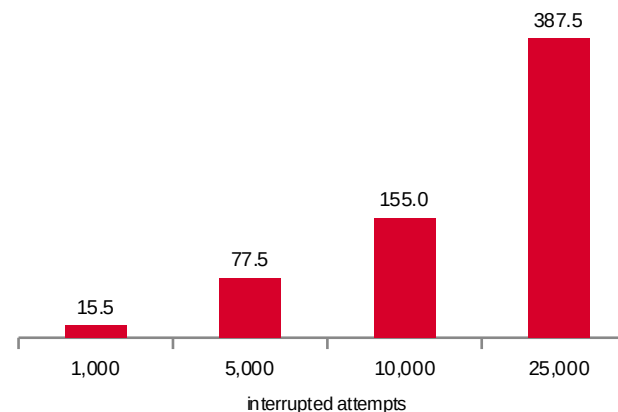
free offline tool for awareness partners; 1930 in every alert



Communities

senior-citizen groups, colleges, CSCs, CA offices

Impact model (illustrative, 2024 I4C averages)



₹1.55 lakh

average loss per digital-arrest case in 2024 (₹1,918 cr / 1,23,672); ₹3.73 lakh in 2025

Every 1,000 interrupted digital-arrest attempts protect about ₹15.5 crore at 2024 averages, before counting tech-support, KYC and investment fraud.



Roadmap: RICE-prioritised, with risks named up front

$$RICE = Reach \times Impact \times Confidence / Effort$$

Next features, ranked by RICE

Feature	Reach	Impact	Conf.	Effort	RICE
 Real-call validation with bank / I4C partners				2 wk	108
 Marathi, Tamil, Telugu, Bengali cue libraries				1.5 wk	74.7
 Family guardian: consented alert to a trusted contact				1.5 wk	53.3
 Same models on Snapdragon 8 Elite phones				4 wk	42.8
 Voice-clone detector on NPU (WavLM, AI Hub)				3 wk	30

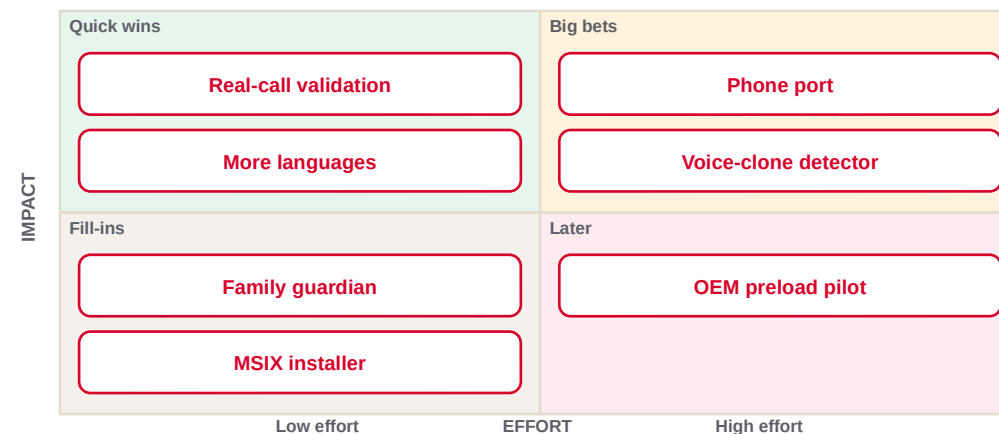
Potential pitfalls and mitigation

-  **False alarms erode trust:** awareness damping, confidence-weighted evidence, hard-negative corpus; 0.5% synthetic false alarms
-  **ASR errors on noisy calls:** screen and system evidence confirm or replace audio; forced language prompt
-  **Scammers change scripts:** tactic-level model; cue library and classifier update offline
-  **Privacy perception:** open source, no network at runtime, nothing stored, verifiable by anyone

Timeline

-  **Now (v1.1)** 3 sentinels, 5 NPU-ready models, fusion, 1930 report, 29 tests
-  **+1 month** on-device benchmark on X Elite, real-call pilot, Marathi
-  **+3 months** voice-clone detector, family guardian, MSIX installer
-  **+6 months** Android on Snapdragon 8 Elite; bank and OEM pilots

Business impact vs effort



THANK YOU

धन्यवाद

Sajag AI turns every Snapdragon AI PC into a private, always-alert guardian against the scams costing Indians thousands of crores a year, and helps victims act in the golden hour.



Code

github.com/tanmaysahare/sajag-ai



Try it

`python -m sajag demo --report`



Author

Tanmay Sahare, IIT Kanpur



If you are being scammed

Call 1930 | cybercrime.gov.in

Qualcomm



Qualcomm AI Hub models on the Snapdragon Hexagon NPU via ONNX Runtime QNN. Built for Snapdragon-powered HP AI PCs.



Appendix A: repository map and how judges can verify

Everything on this deck is reproducible from the repo

Repository map (github.com/tanmaysahare/sajag-ai)

Path	What it contains
sajag/npu.py	ORT session factory: QNN EP, HTP power profiles, context cache, strict mode
sajag/audio/	WASAPI capture, Silero-VAD, NumPy log-mel, tokenizer, AI Hub Whisper loop
sajag/vision/	screen watcher (dHash gate), NPU OCR with static-shape adapter
sajag/signals/	remote-access, call-app and payment-screen detection
sajag/risk/	scam knowledge base, analyzer, classifier, fusion, explainer
sajag/report.py	golden-hour 1930 report: entities + timeline
sajag/ui/	FastAPI + WebSocket dashboard (127.0.0.1)
scripts/	corpus builder, classifier training, NPU OCR prep, evaluation
tests/	29 unit and end-to-end tests
docs/	architecture, NPU optimisation, privacy, evaluation, 156 references
.github/workflows/	CI: tests + demo replay + evaluation on every push

Verify in 5 minutes on any laptop

```
pip install -r requirements.txt
python -m pytest -q           # 29 passed
python -m sajag demo --report  # 5 scenarios + 1930 report
python scripts/evaluate.py     # detection / false alarms
python -m sajag ui             # dashboard, one-click scenarios
python -m sajag bench --out bench.json # NPU vs CPU on a Snapdragon PC
```

Originality and compliance

- ✓ New project built by the participant during the submission period
- ✓ Models from Qualcomm AI Hub and open-source platforms under their licenses (Silero-VAD MIT, Whisper MIT, PaddleOCR / RapidOCR Apache-2.0, Llama 3.2 Community License)
- ✓ Code released under Apache-2.0; no confidential information or trade secrets
- ✓ Demo screens are synthetic and watermarked 'SIMULATED'
- ✓ Icons: Microsoft Fluent Emoji (MIT); logos used only to name products



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